

Projektmunka II.

Hálózatkezelés

Két telephelyes hálózat tervezése, konfigurálása

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Célkitűzés:

Csinszka Csomag informatikai hálózatának megtervezése. Két telephelyes hálózat tervezése és konfigurálása. Az egyik telephelyen (Budapest) internet hozzáférés a másik telephelyen (Gyöngyös) WIFI hozzáféréseinek megvalósítása legyen.



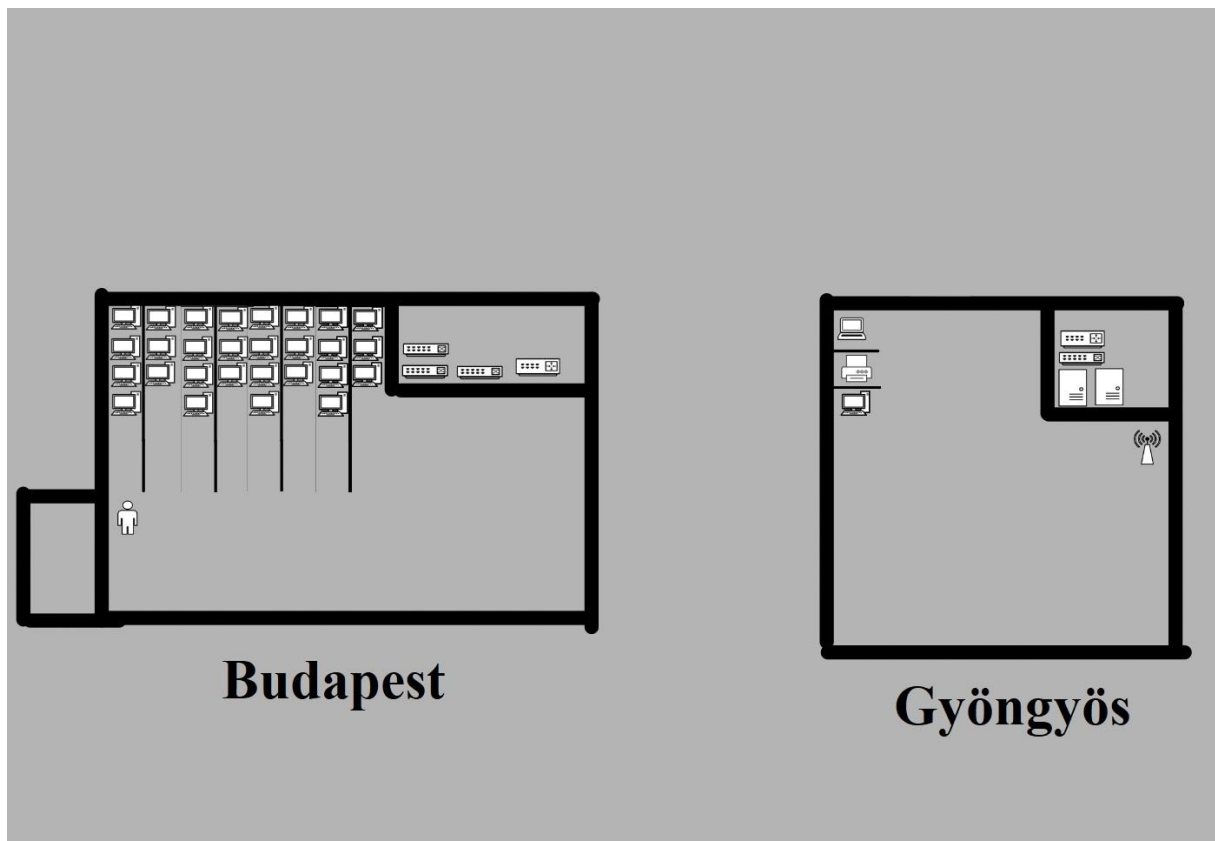
Budapest – Gyöngyös között, egy csomagszállító cég. Mindenféle áruk szállítására.

Eszközök:

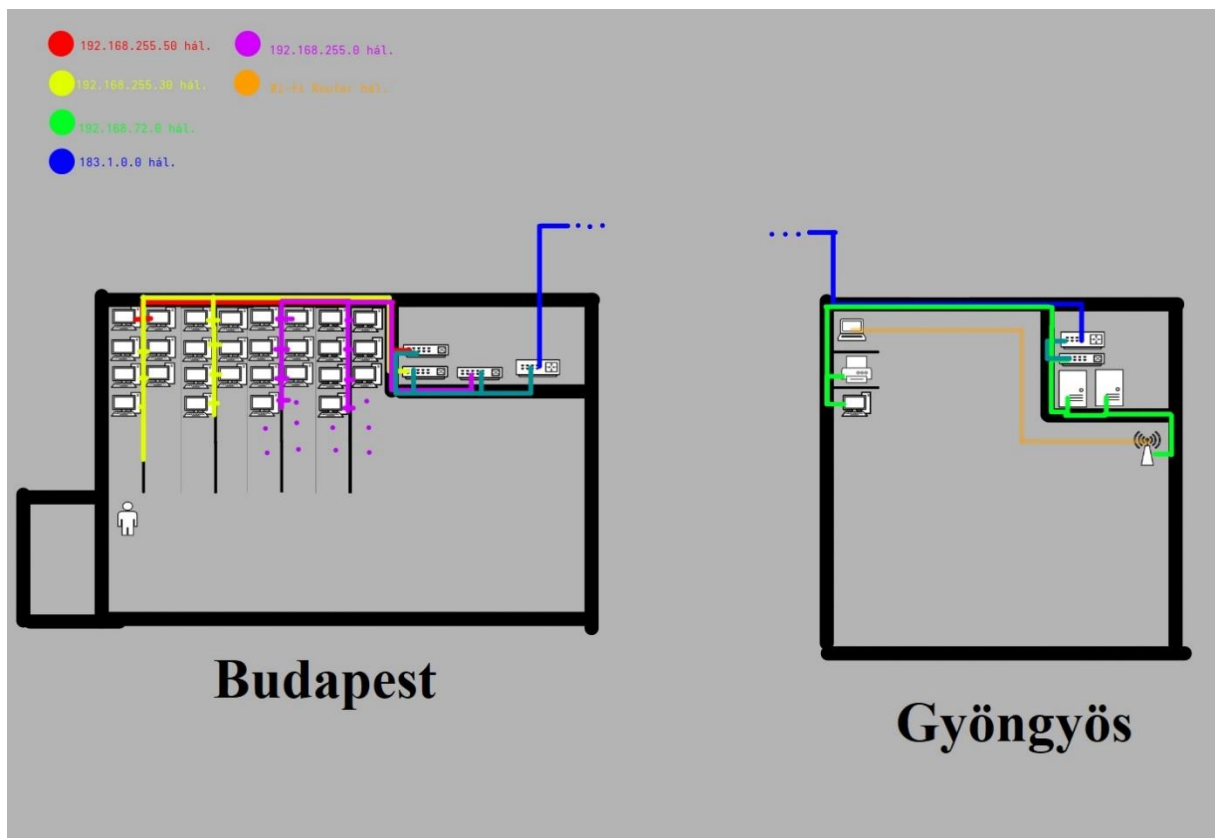
1. Telephely - Budapest	Típus
32 db PC	Dell OptiPlex 7010 Micro (i5, 16GB RAM, 512GB SSD)
3 db Switch	Cisco Business 250-24T-4G (24 portos, Smart Managed)
1 db Router	MikroTik RB5009UG+S+IN

2. Telephely - Gyöngyös	Típus
2 db Szerver	HPE ProLiant DL360 Gen11
1 db Nyomtató	HP LaserJet Pro MFP M428fdw
1 db Laptop	Lenovo ThinkPad L14 Gen 4
1 db Pc	Dell OptiPlex 7010 Micro
1 db WIFI Router (15 kliens hozzáférés)	Ubiquiti UniFi Dream Router
1 db Switch	Cisco Business 250-8T-E-2G (8 portos)
1db Router	MikroTik RB5009UG+S+IN

Fizikai topológia:



Kábelezés terv:



Felhasznált hálózati eszközök:

BUDAPEST

1. TELEPHELY

Eszköz	Interface	IP-cím	Alhálózati maszk	Alapértelmezett átjáró
R1	Gig0/0	192.168.255.33	255.255.255.240	-
	Gig0/1	192.168.255.1	255.255.255.224	-
	Gig0/2	192.168.255.49	255.255.255.248	-
	Se/0/3/0	183.1.0.1	255.255.255.252	-
Switch2	Fa0/2 - Fa0/3	-	-	-
Switch3	Fa0/1 - Fa0/10	-	-	-
Switch4	Fa0/1 - Fa0/20	-	-	-

Eszköz	Interface	IP-cím	Alhálózati maszk	Alapértelmezett átjáró
Hál.1FPC1	-	DHCP – 192.168.255.50	255.255.255.248	192.168.255.49
Hál.1FPC2	-	DHCP – 192.168.255.50	255.255.255.248	192.168.255.49
Hál.1PC1- 10	-	DHCP – 192.168.255.30	255.255.255.240	192.168.255.33
Hál.1PC11-30	-	DHCP – 192.168.255.0	255.255.255.224	192.168.255.1

DHCP kiosztások

POOL_1	Hál.1FPC1- Hál.1FPC2	192.168.255.50-52
POOL_2	Hál.1FPC1- 10	192.168.255.34-44
POOL_3	Hál.1FPC11 -30	192.168.255.2-22

GYÖNGYÖS 2. TELEPHELY

Eszköz	Interface	IP-cím	Alhálózati maszk	Alapértelmezett átjáró
R2	Gig0/0/0	192.168.72.65	255.255.255.248	-
	Gig0/0/1	172.16.0.1	255.255.255.0	-
	Se0/1/0	183.1.0.2	255.255.255.252	-
Switch1	Fa0/1 - Fa0/5	-	-	-
Hál.2Szerv1	Fa0/4	192.168.72.66	255.255.255.248	192.168.72.65
Hál.2Szerv2	Fa0/3	192.168.72.67	255.255.255.248	192.168.72.65
Hál.2Printer	Fa0/2	192.168.72.68	255.255.255.248	192.168.72.65
Hál.2PC1	Fa0/1	192.168.72.69	255.255.255.248	192.168.72.65
Wifi Router	Fa0/0	172.16.0.2	255.255.255.0	172.16.0.1
		10.0.0.1	255.255.255.0	-
Laptop	-	DHCP - kliens	-	10.0.0.1

Wifi Router: 15 kliens hozzáféréssel

WIFI Router

Physical Config **GUI** Attributes

Wireless Tri-Band Home Router Firmware Version: v0.9.7

Setup Setup Wireless Security Access Restrictions Applications & Gaming Administration Status
Basic Setup DDNS MAC Address Clone Advanced Routing

Internet Setup

Internet Connection type: Static IP

Internet IP Address: 172 . 16 . 0 . 2

Subnet Mask: 255 . 255 . 255 . 0

Default Gateway: 172 . 16 . 0 . 1

DNS 1: 0 . 0 . 0 . 0

DNS 2 (Optional): 0 . 0 . 0 . 0

DNS 3 (Optional): 0 . 0 . 0 . 0

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: 10 . 0 . 0 . 1

Subnet Mask: 255.255.255.248

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

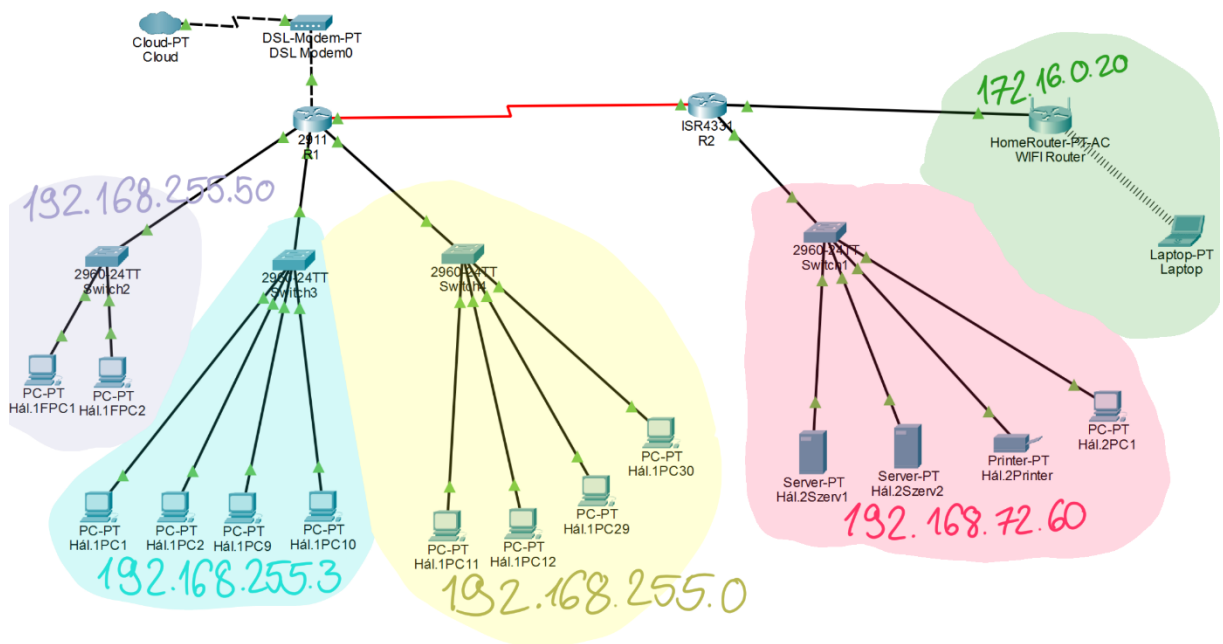
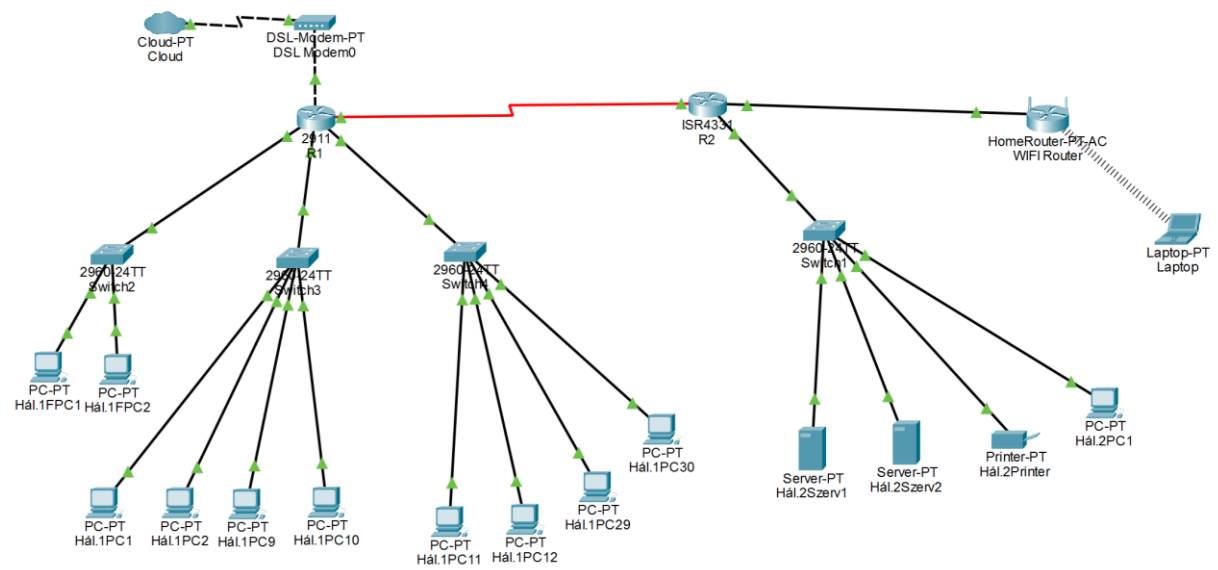
Start IP Address: 10.0.0. 100

Maximum number of Users: 15

IP Address Range: 10.0.0. 100 - 114

Help...

Logikai topológia:



R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets
S 10.0.0.0/24 [1/0] via 183.1.0.2
172.16.0.0/24 is subnetted, 1 subnets
S 172.16.0.0/24 [1/0] via 183.1.0.2
183.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 183.1.0.0/30 is directly connected, Serial0/3/0
L 183.1.0.1/32 is directly connected, Serial0/3/0
192.168.72.0/29 is subnetted, 1 subnets
S 192.168.72.64/29 [1/0] via 183.1.0.2
192.168.255.0/24 is variably subnetted, 6 subnets, 4 masks
C 192.168.255.0/27 is directly connected, GigabitEthernet0/1
L 192.168.255.1/32 is directly connected, GigabitEthernet0/1
C 192.168.255.32/28 is directly connected, GigabitEthernet0/0
L 192.168.255.33/32 is directly connected, GigabitEthernet0/0
C 192.168.255.48/29 is directly connected, GigabitEthernet0/2
L 192.168.255.49/32 is directly connected, GigabitEthernet0/2

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.16.0.0/24 is directly connected, GigabitEthernet0/0/1
L 172.16.0.1/32 is directly connected, GigabitEthernet0/0/1
183.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 183.1.0.0/30 is directly connected, Serial0/1/0
L 183.1.0.2/32 is directly connected, Serial0/1/0
192.168.72.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.72.64/29 is directly connected, GigabitEthernet0/0/0
L 192.168.72.65/32 is directly connected, GigabitEthernet0/0/0
S 192.168.255.0/24 [1/0] via 183.1.0.1